

PROJECT DOCUMENTATION

Sales Intelligence Dashboard

Executive BI asset synthesizing six years of transactional data across Power BI and Tableau.

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ORGANISATION	skynet-datagrid-labs
PLATFORMS	Microsoft Power BI · Tableau Desktop
STACK	DAX · Power Query · LOD Expressions
DATA PERIOD	6 Years (Full Fiscal Cycles)
REPOSITORY	github.com/skynet-datagrid-labs/sales-intelligence-dashboard

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1. Project Overview

This project is a production-grade business intelligence asset built to transform six years of fragmented transactional sales data into a unified executive intelligence layer. The solution is deployed across two leading BI platforms, Microsoft Power BI and Tableau, each serving a distinct analytical purpose while drawing from the same underlying data model.

The Power BI implementation serves as an executive command center, offering multi-page reporting with regional filtering, category-level sales analysis, demographic segmentation, and year-over-year comparisons across 2019 to 2023. The Tableau implementation focuses on strategic analytics, leveraging advanced features including Level of Detail (LOD) expressions, sets, parameters, and map-based geographic analysis to surface insights that go beyond standard charting.

A third component, the Tableau KPI Performance Tracker, delivers state-level sales intelligence across the United States, featuring headline KPI cards, monthly trend lines, and an interactive choropleth map for geographic sales distribution. Together, the three dashboards demonstrate cross-platform BI engineering capability and provide redundancy for enterprise stakeholder preferences.

2. Strategic Objective

Raw transactional records, however voluminous, yield limited strategic insight in their unprocessed form. The core objective of this asset is to close the gap between data collection and actionable intelligence by providing three distinct capabilities to the business:

- **Executive Decision Support:**Real-time visibility into revenue health, regional performance, and customer segment dynamics, delivered through an interface designed for non-technical stakeholders.
- **Profitability Intelligence:**Multi-dimensional analysis across product categories, geographic territories, and temporal trends, enabling granular understanding of where margin is generated and lost.
- **Strategic Planning Foundation:**Historical pattern synthesis across six fiscal years, enabling evidence-based forecasting, seasonality modeling, and resource allocation decisions grounded in data rather than assumption.

3. Dataset Specification

Attribute	Detail
Temporal Coverage	6 years of full fiscal cycles
Data Source	Enterprise transactional database
Record Volume	High-velocity sales transactions
Update Cadence	Real-time synchronization

The dataset is organized across five core analytical dimensions:

Dimension	Description	Analytical Application
Sales Transactions	Line-item revenue records	Revenue trend analysis, product performance
Customer Segments	Demographic and behavioral cohorts	Retention analysis, lifetime value modeling
Geographic Territories	Regional sales distribution	Market penetration, expansion opportunities
Product Categories	SKU-level classification	Portfolio optimization, cross-sell identification
Time Intelligence	Daily, weekly, monthly, quarterly data	Seasonality detection, year-over-year comparison

4. Dashboard Architecture

The solution is structured across five engineering layers, each responsible for a distinct function in the data pipeline from source to visualization:

Layer	Components	Function
Data Ingestion	ETL pipelines, incremental refresh	Automated data extraction from source systems
Data Modeling	Star schema, relationship management	Optimized query performance, logical data organization
Calculation Engine	DAX measures, calculated columns	Complex business logic, time intelligence functions
Visualization Layer	Interactive dashboards, drill-through pages	Executive presentation, exploratory analytics
Security Layer	Row-level security (RLS), role-based access	Data governance, territory-based restrictions

5. Key Performance Indicators

The asset tracks two tiers of KPIs serving different organizational functions:

Executive KPIs

Metric	Definition	Strategic Value
Total Revenue	Gross sales volume (current period)	Overall business health indicator
Revenue Growth	Year-over-year and quarter-over-quarter change	Trajectory assessment
Gross Margin	Revenue minus cost of goods sold	Profitability efficiency
Regional Profitability	Margin contribution by territory	Resource allocation optimization
Customer Lifetime Value	Projected future revenue per customer	Retention investment decisions

Operational KPIs

Metric	Definition	Tactical Application
Average Order Value	Revenue per transaction	Pricing strategy evaluation
Sales Cycle Duration	Lead-to-close timeline	Pipeline efficiency assessment
Win Rate	Closed-won over total opportunities	Competitive positioning analysis
Regional Penetration	Market share by territory	Expansion opportunity identification

6. Technology Stack

Layer	Technology	Purpose
Platform 1	Microsoft Power BI	Executive dashboards, enterprise distribution
Platform 2	Tableau Desktop and Server	Strategic analytics, cross-functional collaboration
Calculation Language	DAX (Data Analysis Expressions)	Complex measures, time intelligence
Data Modeling	Power Query (M language)	ETL, data transformation
Security	Row-Level Security (RLS)	Access governance, territory restrictions
Version Control	GitHub	Asset versioning, collaboration tracking

7. Dashboard Capabilities

Capability	Implementation	Business Value
Real-Time Monitoring	Live data connections, automatic refresh	Immediate anomaly detection
Drill-Down Analysis	Hierarchical navigation: region, territory, account	Root cause identification
What-If Parameters	Dynamic scenario modeling	Strategic planning simulation
Forecasting	Time series decomposition	Predictive revenue planning
Executive Summaries	KPI cards, trend indicators, conditional formatting	At-a-glance performance assessment
Mobile Optimization	Responsive layouts, phone report designer	On-the-go executive access

8. Repository & Access

The full project, including Power BI report files, Tableau workbooks, and all supporting assets, is maintained in the skynet-datagrid-labs GitHub organisation. The repository is version-controlled and open for collaboration.

Resource	Location
GitHub Repository	github.com/skynet-datagrid-labs/sales-intelligence-dashboard
Author	Tony Kenga (ORCID: 0009-0007-6899-8590)
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